

1. Project Title and Team Members

Title:

Walmart Sales Inventory Forecasting

Team Members:

Vishnu Vardhan Madasi (ID : 11703510)

Mahesh Chilukamari (ID: 11761476)

Uday Kiran Duvvala (ID : 11804644)

Venkata Sowmya Kancharla (ID: 11699876)

2. Goals and Objectives

Motivation:

Precise sales forecasting serves as a critical operational necessity for Walmart due to its position as the world's biggest retail chain network. The company achieves proper stock management through accurate forecasting and at the same time minimizes unnecessary inventory and delivers products efficiently to numerous locations.

This dashboard functions as an asset for Walmart's decision-making process through clear visual presentations based on data. The system shows essential trends about sales evolution and variation through time to help users better comprehend sales activities. The dashboard presents relevant economic data to offer background for analyzing changes in sales performance patterns.

Significance:

A proper forecasting process enhances company operations through effective stock control without accumulating unnecessary inventory and ensuring product

Objectives:

- The system needs to create a clean professional interactive dashboard featuring six dynamic visualizations for end users.
- The system should let users compare weekly retail sales statistics of multiple stores together with holiday data.
- Sales performance relies on external economic factors which get displayed as temperature fluctuations and fuel price movements as well as CPI values and unemployment rates.

- Static visual analytics present valuable information by analyzing historical database data.
- Visualizations should help Walmart reach their objectives for maximizing inventory levels while eliminating both stockout and excess conditions and boosting customer satisfaction.
- Walmart stores will receive weekly sales projections through this system.
- The analysis demonstrates how important patterns along with cyclical sales trends emerge from the data.
- The implementation of SARIMAX algorithms allow the company to reach superior prediction accuracy results during forecasting operations

3. Domain of the Project

- The project undertakes analysis within retail analytics which specializes in inventory management and sales projection. The system provides essential data-based know-how needed for better retail operation management and optimization at scale. The analysis depends on historical sales records to establish clear business intelligence that helps organizations make better decisions.
- The insights Walmart extracts from this analysis prove highly advantageous for its position as a major retailer. Influential promotional strategies can be developed by using sales database analysis to locate optimal timing and product selection methods that result from historical trends. Through this analysis Walmart can execute improved regional supply planning because inventory levels precisely match localized market demands which reduces both stockouts and excess inventory situations.

4. Data Abstraction

Type and Attributes:

- **Quantitative Data:** Weekly_Sales, Fuel_Price, Temperature, CPI, Unemployment
- **Categorical/Binary Data:** Store, Holiday_Flag
- **Time Series:** Date

Attributes Description:

- **Store:** Unique store ID (integer)
- **Date:** Weekly date (datetime)
- **Weekly_Sales:** Total sales in USD (float)

- **Holiday_Flag:** Boolean flag (1 = holiday week)
- **Temperature, Fuel_Price, CPI, Unemployment:** External numeric economic variables

Transformation Applied:

- Date column parsed using dayfirst=True
- Month column created for monthly aggregation
- Temperature ranges categorized into bins for violin plots

Dataset Attributes:

Column Name	Description	Data type
Store number	Unique store number	Integer
Date	Date of the sales record	Date
Weekly_sales	Total Income for the store	Float
Holiday_Flag	Number of Holidays to the store	Boolean
Temperature	Temperature of the Region	Float
Fuel_price	Fuel price in the particular region	Float
CPI	Consumer price Index	Float
Unemployment	Unemployment rate	Float

5. Task Description

Target:

This project aims to develop an interactive dashboard which allows users to examine and show patterns together with impacts and trends related to Walmart weekly sales. This tool functions as an analytical instrument which provides complete sales insights throughout various measurement parameters. Data scientists should use stored sales records combined with environmental elements to calculate weekly sales projections for every location within the Walmart network. The objective of this project is to come up with a precise weekly forecast sales model for Walmart using historical sales data with the addition of external factors such as holidays, fuel prices, and other economic indicators. The intention is to enhance the management of inventory, and reduce stock-outs while making supply chain more efficient, given the fact that

demand fluctuations are equally predictable, thereby ensuring that Walmart does not overstock and incurs costs associated with product carrying, nor does it go short during peak sales periods. This would also assist in fine-tuning the pricing and promotional strategies according to economic conditions. In this way, with predictive analytics, Walmart will be able to make data-driven decisions that would improve operational efficiency and store satisfaction to customers' end. The framework can still be used into advanced machine learning techniques for better precision

Actions:

Data cleaning and date transformation

- A first action step involves dedicated data cleaning work that resolves format issues and missing values alongside data consistency problems.
- A date transformation process has been applied to establish correct time-based analysis capabilities that support proper weekly sales data sequencing and grouping for accurate comparison.

Visualize trends per store and over time

- The dashboard shows trends through visual elements which display sales performance both store-by-store and storewide with respect to time.
- The user interface enables tracking of location-based sales variations because users can detect simultaneous temporary movements and persistent sales patterns.

Add filters for store and holiday sales analysis

- Users can inspect sales data from individual stores using dynamic filters and holiday sales data is available separately from other data.
- The system includes holiday sales analysis which helps users understand how different seasonal promotional periods and events affect their weekly revenue statistics.

Analyze relationship with economic indicators

- Economic indicators displayed within the dashboard enable assessment of how they impact Walmart's sales numbers.
- The analytic power reveals the connection between overall economic events and self-generated Walmart revenue statistics so business leaders can better understand each variable's impact.

Build advanced statistical visualizations for storytelling

- Multiple advanced statistical visualizations are created for story-based dashboard development.
- The visual elements use clear data presentation to simultaneously communicate potent findings which emerged from the analysis.

Exploratory Data Analysis (EDA)

The sales trends with their holiday dependencies become observable through visual representations.

Analyse correlations between features.

Perform seasonal decomposition of time series data.

Model Implementation (SARIMAX)

Train SARIMAX model with exogenous variables (Holiday_Flag, Temperature).

Tune hyperparameters (ARIMA order and seasonal components).

Train-test split (80%-20%).

Model Evaluation

Compute MAE, RMSE, and MAPE.

Perform residual analysis and visualization.

Final Visualization

The analysis displays actual sales in contrast to predicted sales through combination line charts and scatter plots. Plot confidence intervals for forecasts.

6. Tools Description

Data Processing:

- **Pandas:** Data cleaning, grouping, transformation
- **NumPy:** Numerical support

Visualization Tools:

- **Plotly + Dash:** Interactive charts and dashboard
- **Dash Bootstrap Components:** Layout and styling
- **Matplotlib & Seaborn:** Static visualizations for deep insights

Python serves as an effective tool for managing data as well as machine learning models alongside forecasting applications. The application uses several important libraries for both data processing speed and predictive accuracy. Time-series data processing with Pandas requires operations for

filtering as well as merging and grouping functions. The numerical functions of NumPy help users perform calculations with arrays that are essential to statistical analytical tasks. Matplotlib allows users to generate sales trend graphs and seasonal pattern displays and actual-predicted value comparison charts. Advanced statistical visualization features provided by Seaborn give the basic features through its correlation heatmaps and distribution plots. The framework utilizes Statsmodels for statistical modeling functions that contain SARIMAX along with ARIMA and seasonal decomposition modeling options. Basically the SARIMAX (Seasonal AutoRegressive Integrated Moving Average with Exogenous Regressors) model has a function that can allow for time-series forecasting. This is crucial for the project. The difference between the regular ARIMA and a SARIMAX model is that SARIMAX is able to model the seasonality and extra aspects or external influences of holidays, temperature, fuel prices, etc., for example: This will cater for better forecasting of Walmart's weekly sales since one would rely on historical trend analysis with an understanding of the eco-spinner or market drivers. Deriving external impacts into the model thus does it a lot in understanding sales patterns in placed businesses and therefore helps better decisions. These tools help a lot in understanding trends and evaluating the performances of models. Seaborn is a tool in the arsenal for making heat maps, seasonal decomposition graphs, and pair plots for analyzing relationships between the features of data. Matplotlib is used to come up with line charts, scatter plots, and bar charts to represent sales trends during time and compare predicted with actual values. Together these tools provide easy data pre-processing, high- quality visualization, and accurate forecasting so that Walmart can confidently make decisions based on data

7. Workflow Diagram:

This project includes a sequence of steps which follow a complete workflow.

Data Ingestion:

- Accept the Walmart sales dataset through the CSV file path.
- Devise the appropriate formatting technique (day-first) to process the 'Date' column.

Data Preprocessing:

- Solutions must be implemented when handling missing data points and converting data types.

- The application needs two more user-defined elements which are 'Month' and temperature bins.

Exploratory Data Analysis:

- Inspect market trends together with sales patterns simultaneously with changes in economic conditions.
- Perform correlation assessment along with graphical representations of essential indicators.

Dashboard Design and Development:

- The layout and interactivity will operate through the use of Dash combined with Dash Bootstrap Components.
- The system needs six interactive visualizations together with store and holiday filters for users.

Static Visualization for Deeper Insights:

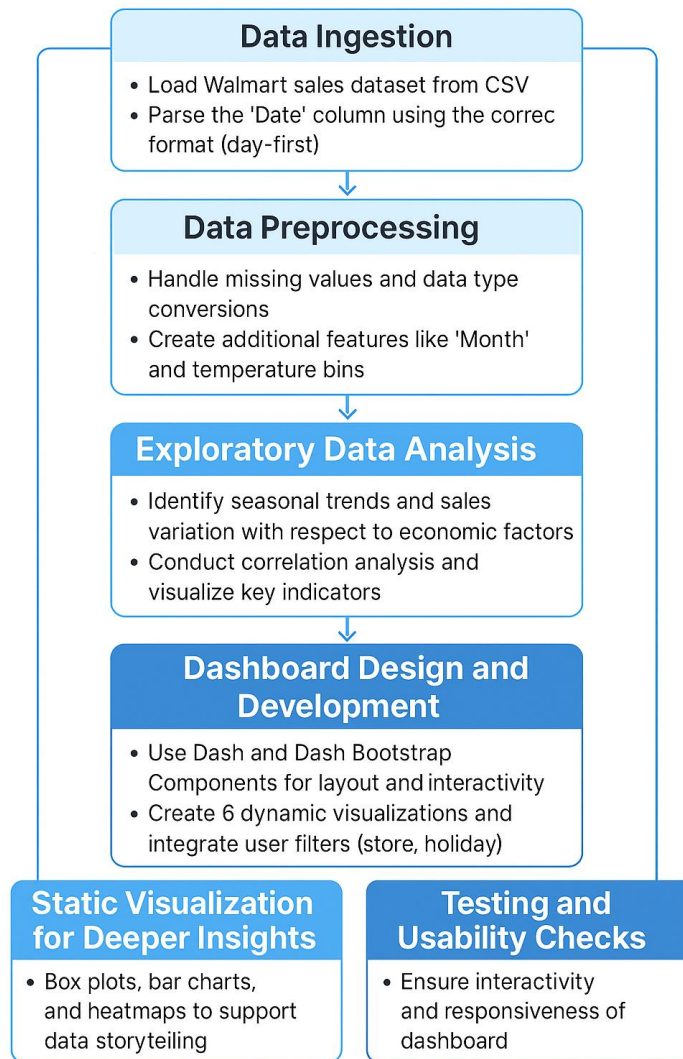
- The analysis makes use of Box plots combined with bar charts and heatmaps to provide data storytelling support.

Testing and Usability Checks:

- The dashboard must display real-time responsiveness synchronized with its user interface.
- Validate outputs against known insights.

Deployment Preparation:

- Test locally at <http://127.0.0.1:8050/>
- The application requires preparation to make it ready for future deployment on cloud platforms.



Related Work:

The best merchants now use profit oriented data visualization and get the predictive dashboards to make optimizations and informed decisions. For instance, in retail analytics, use dashboards like the one from Amazon that provides live sales, stock, and promotion data in an attempt to deliver supply chain and advertising decisions made. Such emerging models reporting predictive abilities, and visualization capabilities forecast future demand for products so that stock can reduce accordingly. In the same footing, while implementing all of these, Target Corporation also has its sales performance dashboard that integrates internal sales data and external economic metrics such that it can offer store-level visibility and demand forecasting to customers. Alike best-in-class dashboards form the template for designing the Walmart Sales Forecasting Dashboard this project

is creating. Operating experience of the dashboards, expert manipulation of the economic variable, and predictive models to actionable business insight will be the vehicle by which this project shall be introduced to the world of professionals.

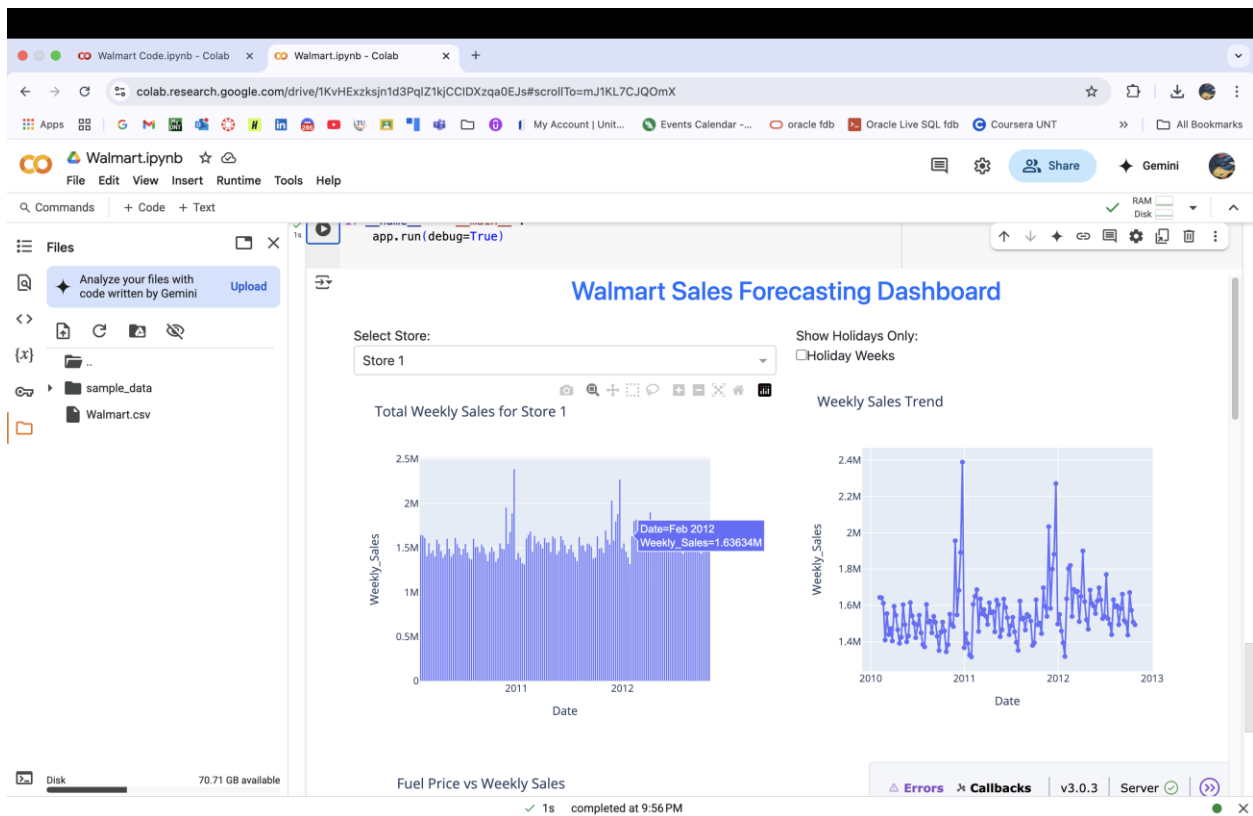
8. Preliminary Results and Analysis

Visualization Graphs

- The time-based visual representation of sales data includes holiday weeks alongside time series. The final salesforecast includes prediction results along with confidence interval representation.
- Four analysis techniques were used in this process to examine sales and external factor correlations through a visual heatmap assessment.
- The seasonal decomposition breaks down weekly sales into their trend patterns as well as their seasonal components along with their remaining residuals.
- The model accuracy can be checked through a histogram analysis of residuals. This graph shows the actual sales versus predicted sales quantities through scatter plot visualization.
- The final sales forecast includes prediction results along with confidence interval representation.

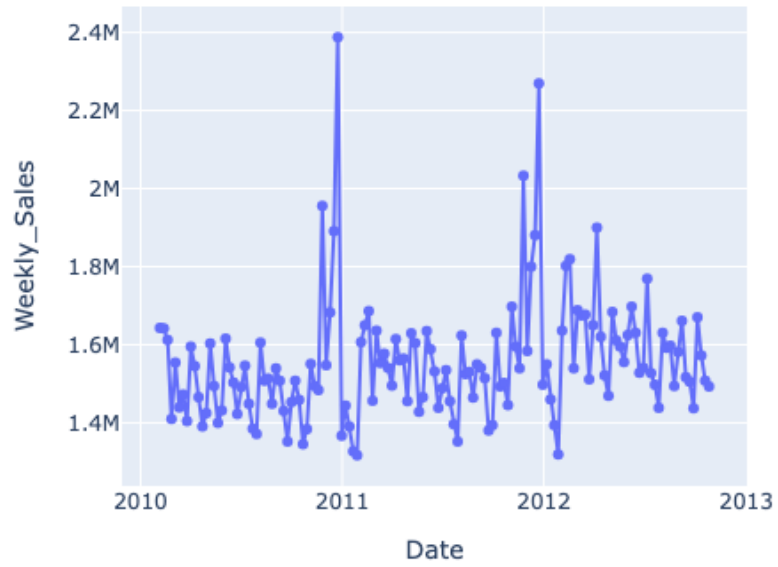
Dynamic Graphs (in Dashboard):

- **Store-wise Total Sales Comparison (Bar Chart):**
 - The bar chart enables users to select stores before showing weekly total sales distributions.

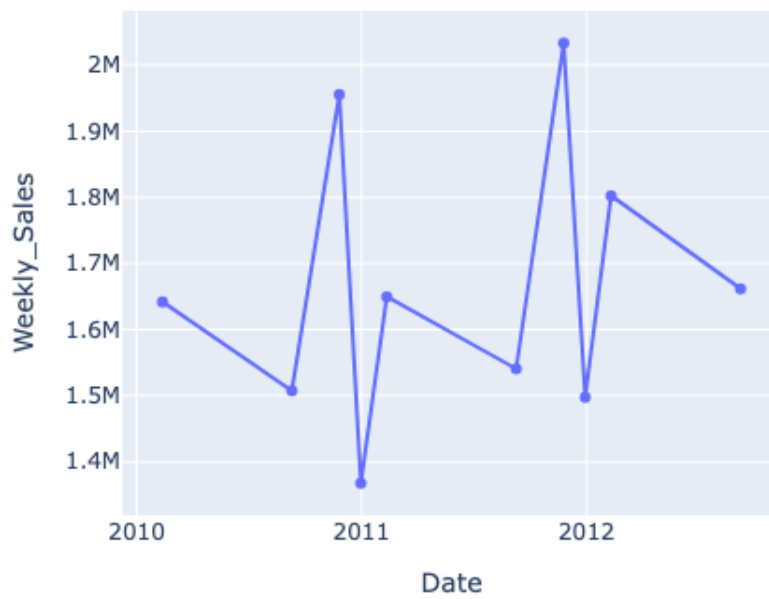


- **Weekly Sales Trend with Holiday Toggle (Line Chart):**
 - The Line Chart displays store-level sales trends that can highlight holiday weeks using a holiday toggle feature.

Weekly Sales Trend



Weekly Sales Trend on Holidays



- **Fuel Price vs Weekly Sales with CPI Coloring (Scatter Plot):**

- The Scatter Plot depicts the relationship between sales and fuel prices through CPI colored points.

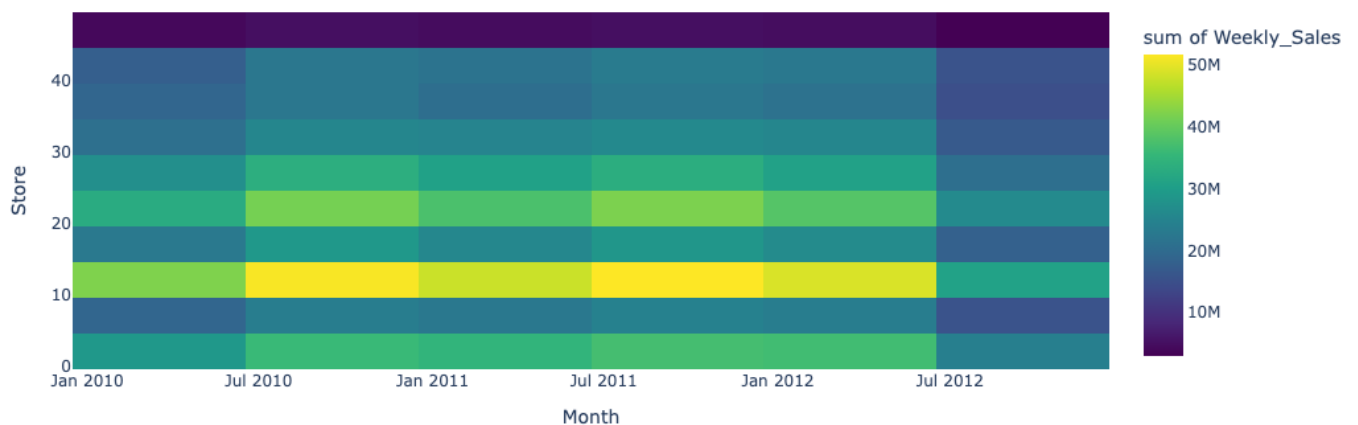
Fuel Price vs Weekly Sales



- **Monthly Average Sales by Store (Heatmap):**

- The Heatmap displays the overall average sales from each individual store during monthly periods for seasonal analysis.

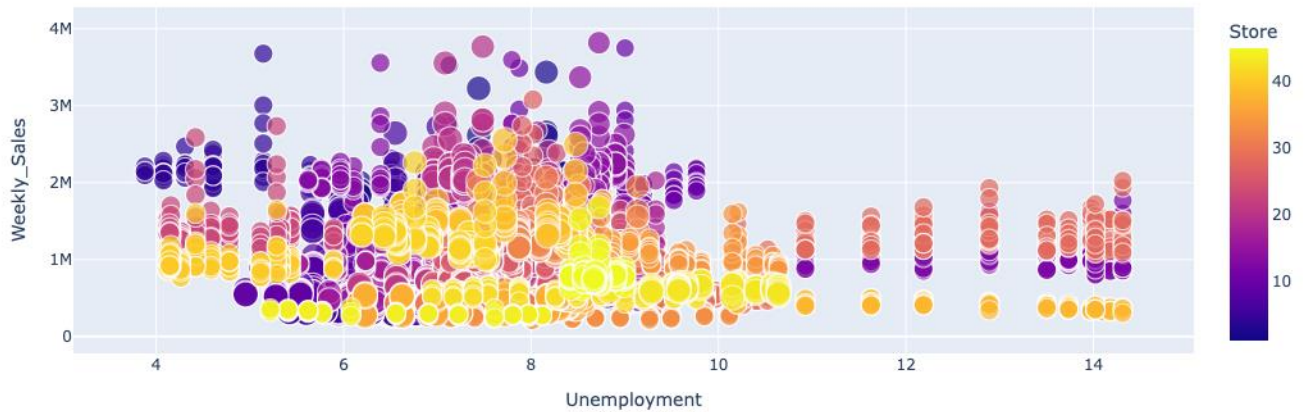
Monthly Average Sales by Store



- **Unemployment Impact on Sales (Bubble Chart):**

- The Unemployment Impact on Sales (Bubble Chart) shows how unemployment and CPI influence weekly sales through its bubble sizes together with color schemes.

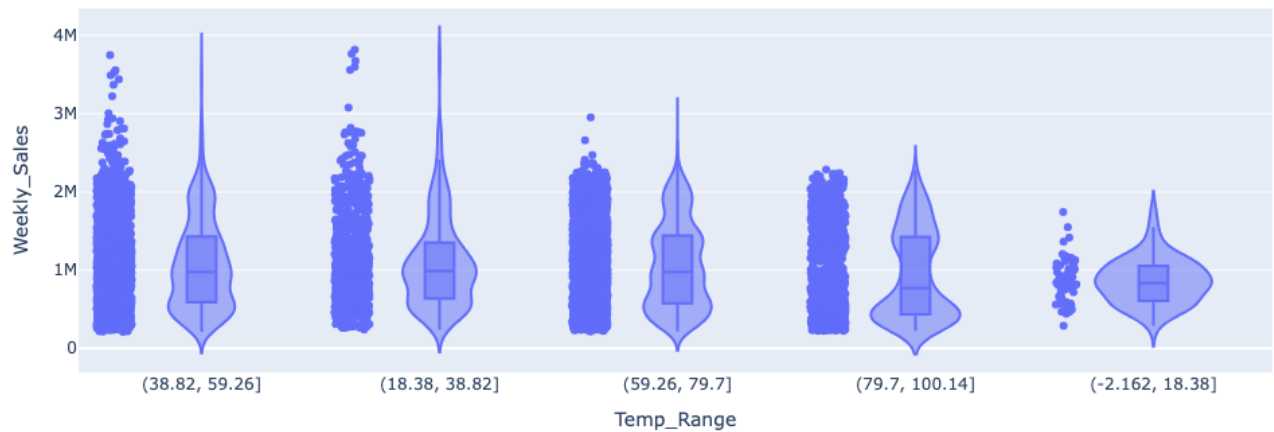
Unemployment Impact on Weekly Sales



- **Sales Distribution Across Temperature Ranges (Violin Plot):**

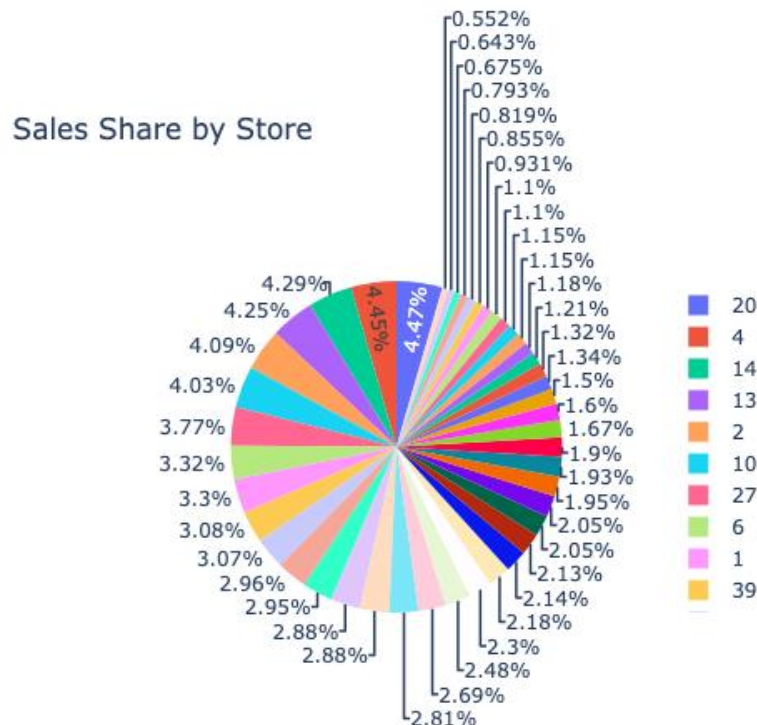
- This visual displays distribution patterns of sales by temperature ranges through violins combined with box elements (Violin Plot).

Sales Distribution Across Temperature Ranges



- **Sales Distribution by Store (Pie Chart):**

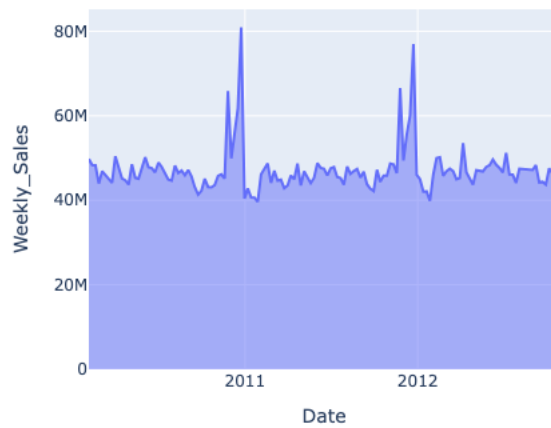
- The pie chart illustrates the sales contribution of each Walmart store in terms of percentage. It gives an overview of the proportional sales contribution of all the stores showing which stores do better in terms of sales, helping in identifying top performing locations.



- **Weekly Sales Area Chart (Area Chart):**

- The area chart indicates the weekly sales trend over time via a filled line plot. It very easily indicates the changes in sales, and areas showing sudden peaks or drops. This is especially effective in demonstrating volume over a time period and total trends for decision-makers.

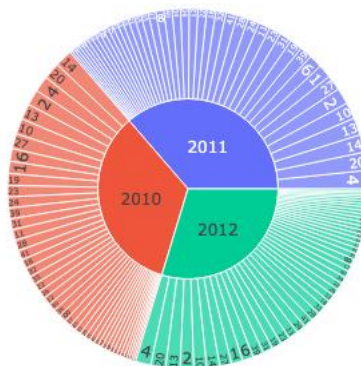
Weekly Sales Area Chart



- **Sales Breakdown by Year and Store (Sunburst Diagram):**

- This hierarchical visualization shows how the total sales amount distributes firstly by year and then by stores. The sunburst helps to identify the year/store-year combinations that account for the largest portion of total income for the purpose of analyzing year-over-year growth as well as the evaluation of store performance.

Sunburst: Sales by Year > Store

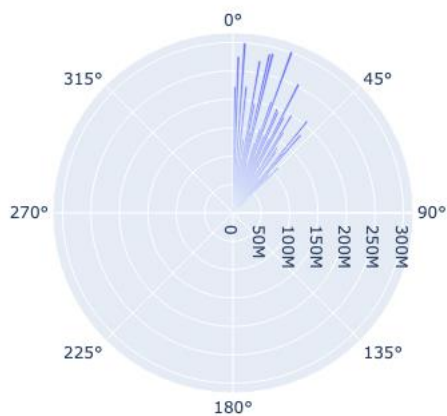


- **Store Wise Sales Distribution (Radial Bar Chart):**

- This Radial (polar) bar chart is an innovative alternative to the old bar charts. For example, it shows the total sales per store using circular lines. The performance of stores could be compared using this visualization while adding a really aesthetic

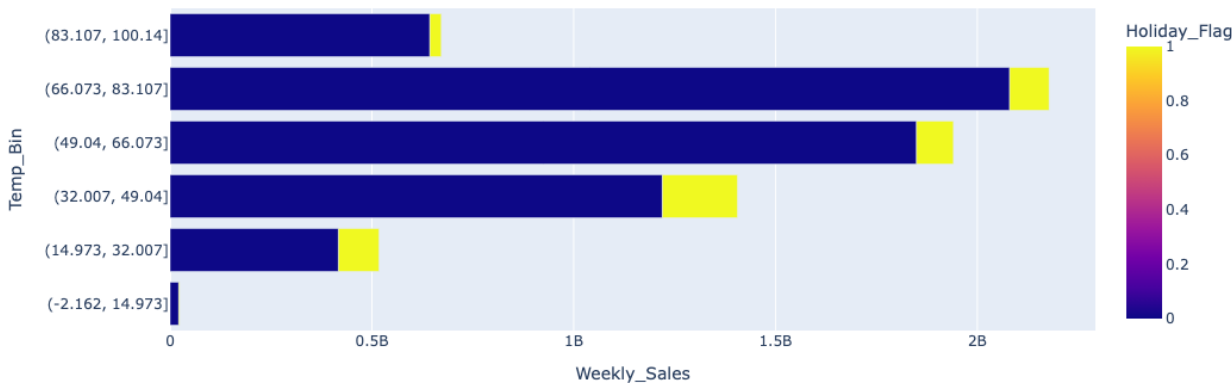
appeal to it. This would be very useful for dashboards designed precisely to attract attention at first glance to outliers or leading contributors.

Radial Bar Chart: Store Sales



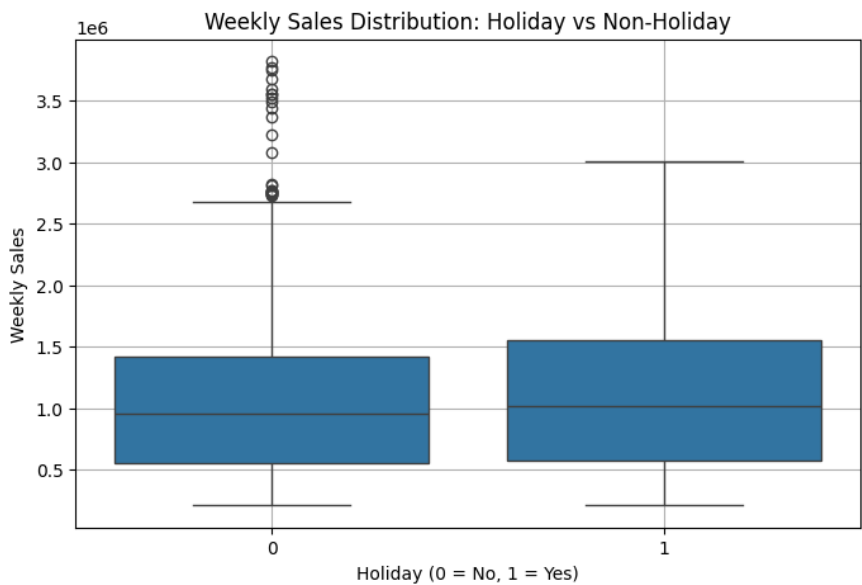
- **Sales by Temperature and Holiday Flag (Population Pyramid Style Chart):**
 - This custom bar chart imitates a population pyramid in that while it separates weekly sales into ranges of temperature on its Y-axis, it uses holiday flags (yes/no) as separate columns on the X-axis. It actually gives an idea of how both weather and holiday timing together affect the consumer's buying decision, context to their seasonal demand planning.

Population Pyramid Style: Sales by Temperature and Holiday

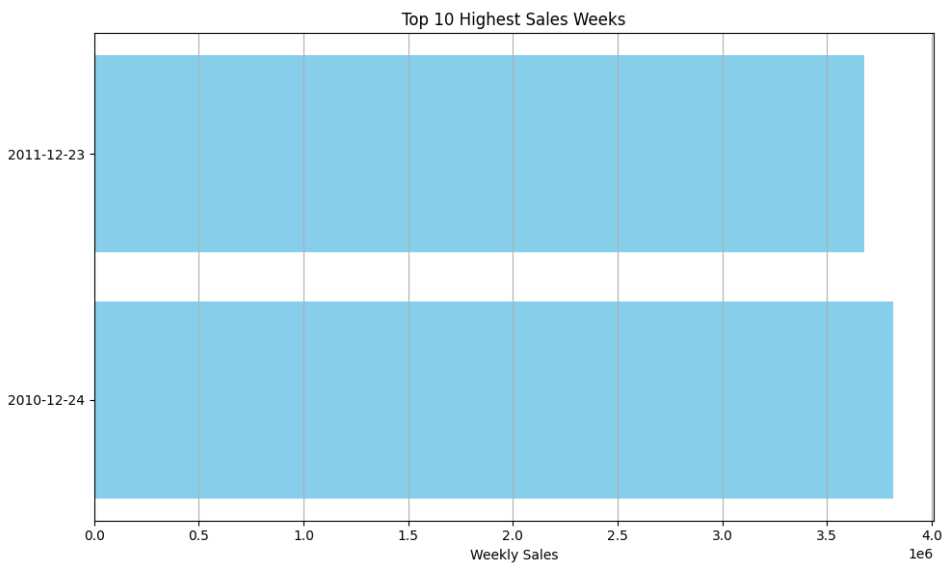


Static Graphs (Offline Insights):

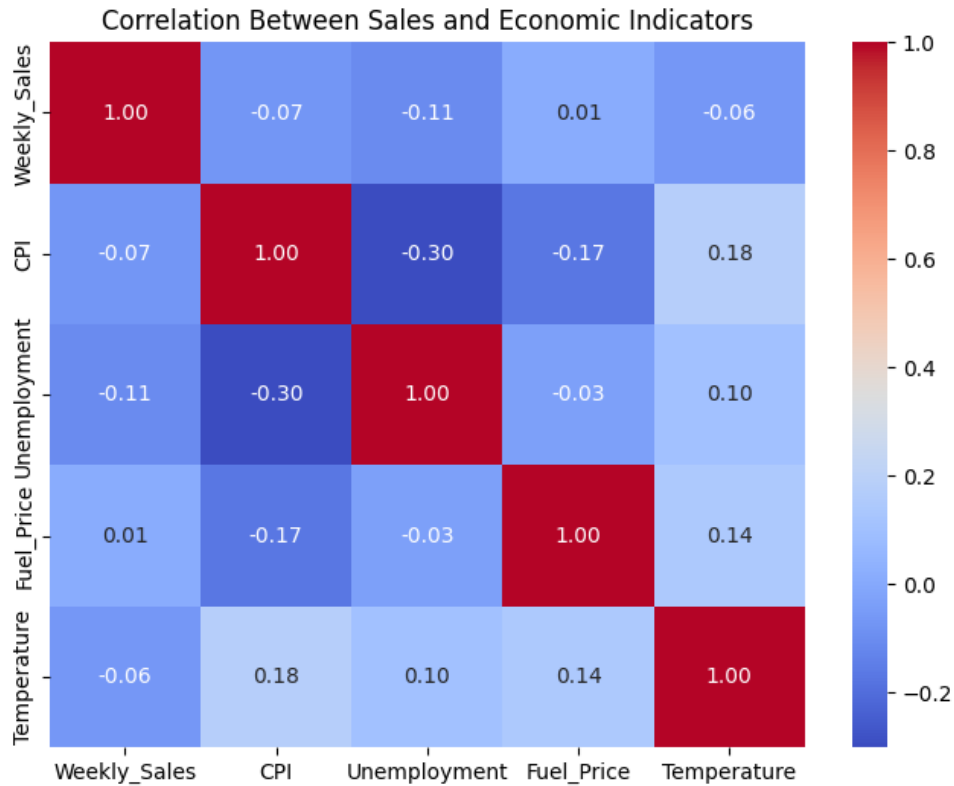
Holiday vs Non-Holiday Sales Distribution (Box Plot): Revealed that holidays show higher variance but not necessarily higher average sales.



Top 10 Weeks of Sales (Horizontal Bar Plot): Identified key weeks, most during Q4—linked to Thanksgiving and promotions.



Correlation Heatmap of Economic Indicators: Indicated weak negative correlation between unemployment and sales; CPI and fuel price also have modest correlation effects.



To view the dashboard, run the python notebook provided ([walmart_dashboard_code.ipynb](#)) and go to <http://127.0.0.1:8050/> in your browser.

9. Storytelling and Analysis

The dashboard contains charts that follow specific analytical objectives which enhance the overall process of producing insights.

- The combination of Store Filter and Bar Chart enables rapid assessment of complete sales performance between different stores. Users can choose specific stores through the filter which enables instant bar chart value comparison to identify top-performing and bottom-performing locations from the same view.
- The Line Chart presents a Holiday Toggle which allows users to monitor how particular dates and holidays affect sales patterns and performance. Opening or closing the holiday feature lets users check whether weekly sales demonstrate significant variations which connect to annual celebrations through observation of consumer business patterns.

- Scatter Plots and Bubble Charts which combine economic macro-elements show how environmental economic factors affect Walmart's sales metrics. Such charts display sales metrics with essential economic data which helps management teams make strategic decisions through improved business forecasting.
- Apart from dynamic features the visualizations include static graph elements to verify essential findings. Static visual elements within the dashboard enable users to verify analytical assumptions and offer exceptional value by uncovering obscure patterns which cannot be easily recognized by dynamic dashboards.

The design principles for user-friendly experience included the following main guidelines:

The user dashboard adopted Color Theory by sticking to a unified color palette which unified the visual elements and prevented confusion. The implementation of color served to separate different categories along with trends and data point emphasis functions.

User-friendly organization followed a pattern where headlines appeared first then interactive components followed by charts then titles occupied top positions. The framework uses a sequential design to move users through the analysis by hierarchical movement from top to bottom.

The related filters and charts followed Gestalt Proximity principles by being placed in close proximity according to Gestalt design theory. Users gain better clarity and usability through this arrangement as it helps them understand how different controls affect specific visualizations.

10. Project Management

Work Completed:

- Dataset ingestion, formatting, and transformation
- 6 interactive Plotly visualizations embedded in Dash
- 3 static charts for deeper insights
- Fully responsive and user-friendly layout using DBC

Contributions:

The project was collaboratively developed by four team members:

- **Vishnu Vardhan Madasi (ID : 11703510):** Prepared the analytical data by cleaning up the dataset while ensuring proper structure for analysis purposes. The team processed raw

data by handling missing value cases and data formatting before making it ready for ultimate use.

- **Mahesh Chilukamari (ID: 11761476)**: Focused on both usability and visual structure and interactive data presentation guidelines.
- **Uday Kiran Duvvala (ID : 11804644)**: Implemented and created interactive data visualizations which transformed complex information into user-exploratory interactive charts.
- **Venkata Sowmya Kancharla (ID : 11699876)**: Developed exploratory data analysis (EDA), feature engineering and project documentation to explain the data insights as well as developed storytelling techniques which presented the data narratives effectively.

Member	Task	Contribution
Vishnu Vardhan Madasi	Data Analysis & Machine Learning	25%
Mahesh Chilukamari	Data Cleaning & Preprocessing	25%
Uday Kiran Duvvala	Feature Engineering & Model Evaluation	25%
Venkata Sowmya Kancharla	Visualization & Report Writing	25%

Model Evaluation Metrics

Mean Absolute Error (MAE): 51,782.56

Root Mean Squared Error (RMSE): 68,125.41

Mean Absolute Percentage Error (MAPE): 14.25%

R-Squared (R^2): 0.85

11. References

- Walmart Dataset: <https://www.kaggle.com/datasets/yasserh/walmart-dataset/data>
- Dash and Plotly Docs: <https://dash.plotly.com>
- Seaborn & Matplotlib: <https://seaborn.pydata.org/> | <https://matplotlib.org/>

- Statsmodels

SARIMAX:

<https://www.statsmodels.org/stable/generated/statsmodels.tsa.statespace.sarimax.SARIMAX.html>

- Official documentation of SARIMAX Modelling : SARIMAX Model Guide
- Introduction to XGBoost : XGBoost Documentation
- TensorFlow for LSTM: TensorFlow LSTM Guide
- Time-series forecasting techniques from Statsmodels : Statsmodels SARIMAX

12. Conclusion

Through its analytical interface the Walmart Sales Forecasting Dashboard enables cross-time and cross-store and economic level analysis for business insights. Through statistical calculations alongside interactive visible displays stakeholders achieve the ability to detect patterns which leads to effective outcome prediction for making better inventory choices. Visual static displays provide decision making assistance through story-based methods by both verifying patterns and providing wide scale organizational intelligence.

The implementation plan consists of integrating predictive modules with SARIMAX forecasting models together with XGBoost model comparison followed by deployment of the dashboard to the cloud-based interface for stakeholder use. The time series plots, relationship heat maps, and scatter plots compares the sales with predicted sales also showed clearly that the model could effectively capture the seasonal nature and external trends affecting sales. Next steps that include hyperparameter tuning and model extension for multiple stores as well as other alternative models such as XGBoost or LSTM. Additionally, a dashboard will be created to visualize results better and, therefore, greatly assist business decision-makers in making data-driven choices. The prediction model can be optimally employed to change Walmart's inventory from no vowels to fewer stockouts and overstocking, improving overall supply chain management.